

SAYAN ROOJ

COMPUTER SCIENCE ENGINEER • AI / ML SYSTEMS ARCHITECT

Email: sayan.rooj@futureverse.ai | Phone: +91 98765 43210 | Location: Bengaluru, India | GitHub: github.com/sayanrooj | LinkedIn: linkedin.com/in/sayanrooj | Portfolio: FUTUREVERSE Platform

PROFESSIONAL SUMMARY

Accomplished **Computer Science Engineer (CSE)** and **AI Systems Architect** specializing in end-to-end Machine Learning pipelines, LLM orchestration, agentic architectures, and high-performance full-stack web platforms. Creator and Lead Architect of **FUTUREVERSE**, an enterprise-grade AI recruitment platform featuring 100% weighted criteria evaluation, automated NLP resume matching, and adaptive proctored video interviews. Proven track record transforming cutting-edge AI algorithms into robust, production-hardened software solutions.

TECHNICAL COMPETENCIES

AI & Machine Learning:	Deep Learning, NLP, Transformers, LLM Fine-Tuning, RAG Architectures, PyTorch, TensorFlow, Scikit-Learn, Hugging Face, LangChain, Semantic Embeddings
Full-Stack & Backend:	Python (FastAPI, AsyncIO, Flask), TypeScript, React 19, Node.js, REST APIs, WebSockets, Tailwind CSS, Responsive UI Design, Vite
Databases & Storage:	PostgreSQL, SQLite (Async SQLAlchemy), Redis, Vector Databases (Pinecone, ChromaDB, FAISS), Alembic Migrations
Cloud, DevOps & MLOps:	Docker, Kubernetes, AWS (EC2, S3, SageMaker, Lambda), CI/CD Pipelines, Git, Model Serving & Optimization, Linux/Bash
Engineering & Security:	System Architecture, Microservices, RBAC Governance, OAuth2, JWT Bearer Security, Anti-Cheating Telemetry, TDD

KEY ENGINEERING PROJECTS & ARCHITECTURES

FUTUREVERSE — AI-Powered Recruitment & Talent Intelligence Platform

Lead Architect & Full-Stack Developer

- Architected an enterprise full-stack platform transforming hiring via explainable AI screening and human-in-the-loop decision governance.
- Engineered a granular 12-section Job Requirement Builder and strict 100% criteria weighting validator with automated historical snapshot auditing.
- Implemented NLP-driven resume parsing with TF-IDF and semantic similarity algorithms, producing verifiable evidence badges (Met, Not Met, Partially Met).
- Developed a proctored AI interview environment featuring camera/mic telemetry, real-time tab switch & blur detection, and automated skill-gap analysis.
- Constructed role-based dashboards (Super Admin, Recruiter/Owner, Candidate) with real-time audit logging and recruitment pipeline analytics.

Autonomous Neural Agentic RAG & Knowledge Synthesizer

Lead AI Engineer

- Built an autonomous multi-agent Retrieval-Augmented Generation (RAG) system utilizing hybrid dense/sparse vector embeddings and rerankers.
- Achieved a 94.2% semantic accuracy benchmark across complex technical documentation with self-correcting query decomposition agents.
- Containerized inference endpoints using FastAPI and Docker, achieving sub-120ms token latency under high concurrency.

Real-Time Edge Vision Proctoring & Integrity Telemetry

Computer Vision Architect

- Designed lightweight client-side computer vision telemetry using MediaPipe and WebAssembly to track eye gaze, head pose, and face absence.
- Interfaced with WebSocket streaming servers to flag suspicious assessment behavior with zero latency and minimal client overhead.

EDUCATION

Bachelor of Technology (B.Tech) — Computer Science & Engineering

Graduation: 2026

Specialization: Artificial Intelligence & Machine Learning | **Standing:** First Class with Distinction (GPA: 9.2 / 10.0)

Core Coursework: Data Structures & Algorithms, Deep Learning, Natural Language Processing, Distributed Systems, Operating Systems, Database Management Systems, Cloud Computing.

CERTIFICATIONS & KEY ACCOLADES

- Deep Learning Specialization** — DeepLearning.AI / Coursera (Neural Networks, CNNs, Sequence Models)
- AWS Certified Cloud Practitioner / Solutions Architect** — Cloud Infrastructure & Scalable Systems
- First Place / National Finalist** — National Level AI & Full-Stack Innovation Hackathons (2024 & 2025)
- Open-Source Contributor** — Active contributions to modern AI tooling, FastAPI extensions, and developer utilities.